Regex-based Log Forensic Engine

System reliability engineers analyze massive volumes of unstructured log data to diagnose service failures and identify operational trends. High-performance forensic tools utilize complex Regular Expressions (regex) with named capture groups to transform raw text into structured datasets for aggregate reporting and targeted anomaly detection.

Implement a `LogForensics` class that utilizes a single, optimized regex pattern to parse standardized system logs. The engine must extract timestamps, log levels, originating components, and messages, while providing analytical summaries and error-specific filtering for post-mortem investigations.

Example 1 Input:

```
LOG 2023-10-27 10:15:30 [ERROR] DatabaseModule: Connection failed
STATS
```

Output:

```
ERROR: 1
```

Explanation: A raw log line is processed. The engine's regex identifies the level as `ERROR`. The `STATS` command then aggregates counts for all discovered log levels.

Example 2 Input:

```
LOG 2023-10-27 10:15:35 [INFO] AuthModule: User logged in REPORT
```

Output:

```
AuthModule: 1
```

Explanation: The forensic engine identifies the originating module as `AuthModule`. The `REPORT` command provides a frequency distribution across all system components.

Function Parameter:

- line (str): A raw string from a system log file.

- component_name (str): Target module for error filtering.

Returns

- dict: Statistical summaries by log level or component.
- list: Error messages associated with a specific module.

Constraints

- Utilize named capture groups: `(?P<timestamp>...)`, `(?P<level>...)`, `(?P<component>...)`, `(?P<message>...)`.
- Log Format: `YYYY-MM-DD HH:MM:SS [LEVEL] Component: Message`.
- Unstructured lines that do not match the pattern must be ignored.

Input Format for Custom Testing Input from stdin contains commands: `LOG <line>`, `STATS`, `REPORT`, or `ERRORS <component>`.

Sample Case 0 Sample Input

```
LOG 2023-10-27 12:00:00 [ERROR] Network: Timeout ERRORS Network
```

Sample Output

```
ERROR: Timeout
```

Explanation The system identifies an error in the Network component and retrieves the associated message.